RBF-SAM: Robust Building Footprint Extraction From Off-Nadir Remote Sensing Images via Segment Anything Model

Yuxuan Li, Keming Chen, *Member, IEEE*, Zicheng Lei, Yating Yang, Jinsong Lv, and Zhi Zhou

***Abstract*—Extracting building footprints from off-nadir imagery has emerged as a fundamental task in geospatial object detection, offering an accessible and efficient solution for large-scale urban mapping. Recently, the segment anything model (SAM) has demonstrated strong capabilities for such tasks across diverse remote sensing settings. However, its performance remains sensitive to prompt quality, which, in turn, constrains practical deployment when available prompts are variable or suboptimal. To address this challenge, we propose the robust building footprint SAM (RBF-SAM), an extraction framework tailored for off-nadir remote sensing imagery. Specifically, guided by a prompt-robust training (PRT) strategy, RBF-SAM effectively alleviates prompt sensitivity across three progressive levels: prompt encoding, prompt–image interaction, and image representation. To handle variations in input prompts at the encoding stage, we introduce a prompt adaptation encoding (PAE) module to generate category-aware representations. During the prompt–image interaction stage, a multiscale shift-aware attention refinement (MSAR) module is embedded, driven by the prompt representations to dynamically correct attention shifts and feature misalignments. Furthermore, to capture fine-grained details, we design a wavelet-based frequency enhancement (WFE) module that exploits frequency-domain information for offset prediction and boundary delineation, enabling accurate and stable instance-level building footprint extraction. Experiments conducted on three public datasets [i.e., Buildings in Off-Nadir Aerial Image (BONAI), OmniCity-view3, and Huizhou] indicate that RBF-SAM remains robust across varying prompt types and qualities while achieving competitive results. Under the optimal prompt setting, RBF-SAM outperforms the state-of-the-art PolyFootNet, with *F*1-score improvements of 7.09%, 1.89%, and 4.97%, respectively. The code of RBF-SAM will be available at https://github.com/Xuaner624/RBF-SAM**

***Index Terms*—Building footprint extraction, instance segmentation, model robustness, off-nadir remote sensing images, segment anything model (SAM).**

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I. Introduction

ACCURATE building footprint extraction from remote sensing imagery serves as a fundamental basis for analyzing urban spatial structures, assessing living environment quality, and supporting smart city development [1], [2], [3]. In recent years, this task has made significant progress with the adoption of deep learning paradigms, as these methods can autonomously learn discriminative feature representations, gradually replacing traditional approaches primarily reliant on hand-crafted features [4], [5], [6], [7]. However, most existing extraction methods are designed for near-nadir remote sensing imagery, where the roof and footprint projections of buildings are largely overlapped. In contrast, off-nadir imagery comprises a substantial portion of remote sensing image archives, enables flexible acquisition conditions, and provides complementary facade structural information, holding significant potential for practical applications [8], [9], [10]. Therefore, achieving accurate and robust building footprint extraction in off-nadir scenarios remains a critical challenge in remote sensing interpretation.

Specifically, the oblique viewing angle inherent in off-nadir imagery leads to significant roof-to-footprint projection displacements, often exacerbated by facade occlusions and shadow effects [14], [15], [16]. To address these challenges, existing studies primarily adopt offset learning as a core design paradigm, leveraging the structural consistency between building roofs and their corresponding footprints, with representative methods including LOFT [17] and MLS-BRN [18]. Rather than directly segmenting the occluded ground regions, these approaches [19], [20], [21], [22] predict more visible roof contours along with spatial offset vectors to infer the true footprints, achieving significant performance improvements. Recently, foundation models such as the segment anything model (SAM) [11] have reshaped remote sensing interpretation, owing to their strong zero-shot generalization and semantic understanding [23], [24], [25]. Building upon the offset-based framework, methods such as OBM [13] and PolyFootNet [26] leverage SAM to handle off-nadir scenarios, effectively breaking through the accuracy limitations and generalization constraints of previous approaches.

However, the exploration of SAM for off-nadir building footprint extraction is still at an early stage. Current

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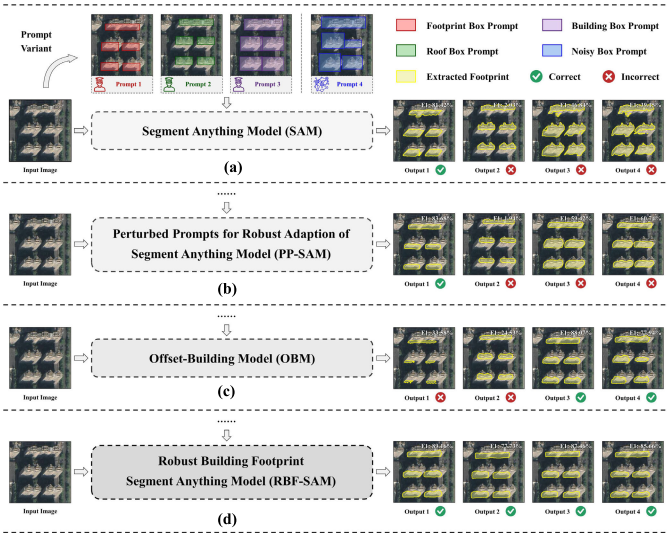


Fig. 1. Prompt sensitivity of SAM-based methods for off-nadir building footprint extraction. We compare the segmentation results of SAM [11], PP-SAM [12], OBM [13], and the proposed RBF-SAM under different types of box prompts. Both the original SAM and existing adapted variants, such as the prompt-robust PP-SAM and the task-specific OBM, remain sensitive to prompt variations in off-nadir scenes, showing performance fluctuations under noisy prompts. In contrast, RBF-SAM maintains consistent and accurate footprint extraction across diverse prompting conditions, demonstrating superior robustness. Corresponding $F1$-scores are annotated in the upper-right corner. (a) General foundation framework. (b) General prompt-robust framework. (c) Off-nadir task-specific framework. (d) Our off-nadir prompt-robust framework.

approaches [13], [26] are mostly optimized for high-quality prompts, typically derived from predefined ground-truth bounding boxes. Nevertheless, the model's sensitivity to prompt uncertainty remains a major limitation in practical deployment [12], [27], [28]. This issue becomes particularly pronounced under complex off-nadir imaging scenarios, where obtaining reliable prompt boxes faces both objective and subjective challenges [29], [30]. As illustrated in Fig. 1, due to the inherent geometric ambiguities of oblique views, even expert annotators adhering to standard protocols may define bounding boxes differently, based on the roof, the footprint, or the overall building structure. Although these annotations are all semantically valid, the original SAM primarily responds to the prompted regions, resulting in noticeable output discrepancies and segmentation errors under prompt variations. More critically, even current prompt-robust and task-tailored frameworks struggle to overcome this limitation. For instance, PP-SAM [12] provides certain performance improvements, yet remains sensitive to prompt-target misalignment. Similarly, while OBM [13] performs well under its training conditions (i.e., ground-truth box prompts), it still fails to generalize to diverse prompts. Furthermore, this performance degradation is amplified by noisy bounding boxes from crowdsourcing platforms and inaccurate annotations from nonexpert users [31]. Unfortunately, to date, no sufficiently effective solution has been proposed to adapt SAM to such diverse prompting conditions for off-nadir building footprint extraction.

In this study, we introduce the robust building footprint SAM (RBF-SAM), a novel framework for accurate and robust building footprint extraction from off-nadir imagery, specifically tailored to handle prompts of varying types and qualities. To achieve this, RBF-SAM employs a prompt-robust training (PRT) strategy that optimizes the architecture across the prompt encoding, prompt–image interaction, and image representation levels. First, to accommodate diverse prompt types derived from distinct building annotations (e.g., roof and footprint), a prompt adaptation encoding (PAE) module is incorporated at the prompt encoding level. Subsequently, a multiscale shift-aware attention refinement (MSAR) module is introduced at the prompt–image interaction stage to alleviate attention shifts and feature misalignments induced by inaccurate prompts. Finally, a wavelet-based frequency enhancement (WFE) module refines offset predictions in the frequency domain and integrates spatial–frequency features to strengthen representation quality, thereby ensuring accurate and stable footprint extraction. Extensive experiments on three public datasets demonstrate that RBF-SAM achieves significant improvements in prompt robustness and cross-region generalization compared with state-of-the-art methods. Our main contributions are summarized as follows.

1) We propose RBF-SAM, a robust SAM-based framework for off-nadir building footprint extraction. To the best of our knowledge, it is the first framework to address segmentation instability under diverse prompt conditions in such complex scenarios.
2) We develop a prompt-robust architecture integrating three novel modules: PAE, MSAR, and WFE, jointly optimizing prompt encoding, prompt–image interaction, and image representation for stable footprint extraction.
3) We validate the robustness and generalization of RBF-SAM on the Buildings in Off-Nadir Aerial Image (BONAI), OmniCity-view3, and Huizhou datasets. Furthermore, we explore the impact of different prompts on SAM-based networks within offset paradigms, providing insights for future off-nadir building extraction.

II. Related Work

A. SAM and Its Applications

The SAM [11] stands as a pioneering foundation model for image segmentation. Built upon the SA-1B dataset containing over 1.1 billion masks, SAM introduces a prompt-driven interactive segmentation paradigm and demonstrates strong zero-shot generalization capability across diverse visual scenarios. Since its introduction, numerous studies have explored extensions and adaptations of SAM. HQ-SAM [32] introduces learnable high-quality tokens to enhance mask prediction accuracy while preserving generalization capability, whereas BA-SAM [33] addresses spatial scale discrepancies among datasets to improve adaptability across different image resolutions. To meet the efficiency demands of real-world deployment, lightweight variants such as FastSAM [34], MobileSAM [35], and EfficientSAM [36] have been proposed, which reduce computational overhead through structural simplification or knowledge distillation frameworks. Concurrently, SAM-Adapter [37] and SAMed [38] adopt parameter-efficient fine-tuning mechanisms, including adapter modules [39] and low-rank adaptation (LoRA) [40], to explore the transfer of SAM's rich representations and generalization capabilities to specific downstream tasks.

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In the realm of remote sensing, SAM has also demonstrated significant potential. SAMRS [41] utilizes ground-truth bounding boxes from existing detection datasets as prompts to construct a large-scale remote sensing segmentation dataset. To alleviate the reliance on manual interaction, RSPrompter [24] learns prompt embeddings to enable fully automated instance segmentation in remote sensing imagery. MeSAM [42] explores an efficient fine-tuning paradigm to better adapt SAM to remote sensing semantic segmentation tasks. RingMo-SAM [23] further extends SAM's foundational modeling capabilities to multimodal data, covering both optical and synthetic aperture radar (SAR) imagery. Furthermore, focusing on geospatial object recognition tasks, RSAM-Seg [43] incorporates high-frequency information to fine-tune SAM, leading to notable improvements in cloud, building, and road extraction performance. Similarly, UrbanSAM [25] models multiscale invariance to achieve accurate extraction of urban elements, including buildings, roads, and water bodies. However, research on SAM-based approaches for off-nadir building footprint extraction remains limited. The few existing works primarily involve straightforward adaptations of SAM without task-specific considerations, underscoring the need for further investigation [13], [26].

B. Building Footprint Extraction From Off-Nadir Imagery

Building footprint extraction has been extensively investigated in near-nadir remote sensing imagery, leading to a series of notable advances [4], [5], [7], [44], [45]. In contrast, off-nadir imagery introduces substantial challenges for accurate footprint delineation, owing to significant roof-to-footprint projection misalignment, as well as severe shadowing and facade occlusions. To address these challenges, mainstream approaches typically rely on offset-based modeling by leveraging the structural consistency between roofs and footprints. Early studies mainly focused on pixel-level segmentation frameworks. Christie et al. [8] proposed a U-Net-based framework that jointly predicts image-level orientation along with pixel-wise height values and magnitudes, from which flow vectors are derived to warp building annotations to footprints. MTBR-Net [14] further models the visible parts of buildings and their spatial relationships, and incorporates prior knowledge for optimization, resulting in improved building footprint extraction accuracy. Building upon this, WS-MTBRNet [20] introduces a weakly supervised learning strategy and achieves competitive performance even with limited annotated samples.

Subsequently, an increasing number of studies have reformulated off-nadir building footprint extraction as an instance segmentation problem. As a pioneering work in this direction, LOFT [17] builds upon Mask R-CNN [46] to jointly predict roof masks and roof-to-footprint offsets, thereby addressing the misalignment between roofs and footprints. MLS-BRN further proposes a multilevel supervision strategy that flexibly leverages samples with different annotation granularities, enabling cost-effective footprint extraction [18]. BDH-Net [21] integrates Mask2Former [47] with an interinstance attention graph to achieve reliable extraction of building footprints and heights. Furthermore, Shao et al. [19] improved the accuracy of roof extraction and offset estimation by introducing a cross-field auxiliary task and a multiscale mask head, while DG-BRF [22] introduces diffusion models and geometric priors to further enhance extraction accuracy. More recently, OBM [13] pioneered the application of SAM to off-nadir building extraction tasks. Building upon this, PolyFootNet [26] introduces a self-offset attention mechanism to enable polygonal building footprint extraction. DragOSM [30] further explores the correction of spatial discrepancies between historical OpenStreetMap labels and current imagery. While the aforementioned methods have achieved promising results, they predominantly operate within the spatial domain, leveraging multiscale aggregation or attention mechanisms to enhance representation learning. Despite the proven effectiveness of frequency-domain representations in various visual tasks [48], [49], [50], their potential remains underexplored in off-nadir building extraction, particularly for correcting roof-to-footprint misalignment and enhancing fine-grained footprint delineation.

C. Robustness of SAM

Currently, research on the robustness of SAM remains relatively limited. Most existing works concentrate on evaluating its performance under various adversarial conditions or corruption scenarios, providing a preliminary understanding of its capability to handle real-world challenges [51], [52], [53]. Beyond these analyses, RobustSAM [54] mitigates the performance decline caused by image degradation by enforcing a consistency constraint that aligns representations of degraded images with those of high-quality images. However, such approaches primarily investigate robustness at the image level, while overlooking the pivotal role of prompts in stability assessment and real-world deployment. Consequently, they largely depend on high-quality manual prompts to ensure precise segmentation. Research by [13] and [27] reveals that SAM is highly sensitive to prompt uncertainty, particularly under box prompt settings, where performance degradation becomes increasingly severe as the shift magnitude increases. To enhance prompt-level robustness, PP-SAM [12] employs a simple fine-tuning strategy using variable box prompts, effectively improving the model's resilience to box prompt perturbations. Stable-SAM [31] addresses the attention shift induced by low-quality prompts by calibrating the cross-attention mechanism via adjustments to the sampling locations and amplitudes of image features, thereby improving segmentation stability across varying prompt qualities. Furthermore, RoBox-SAM [28] targets low-quality box prompts by introducing a refinement module to explicitly rectify box coordinates and integrating autogenerated point prompts with a self-information mining strategy to comprehensively enhance robustness. However, current prompt-level approaches mainly focus on natural or medical scenes, where prompts are generally aligned with the target regions. In contrast, off-nadir building footprint extraction presents more complex conditions: the target footprint is often occluded, and prompts originate from diverse spatial references (e.g., roofs, buildings, footprints, or inaccurate annotations), typically leading to severe prompt–target spatial misalignment. Under such conditions, conventional direct prediction paradigms struggle to achieve robust footprint extraction due to the lack of explicit spatial relationship modeling, such as roof-to-footprint offsets.

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TABLE I

COMPARISON OF REPRESENTATIVE SAM-BASED FRAMEWORKS FOR PROMPT ROBUSTNESS IN OFF-NADIR REMOTE SENSING IMAGERY

Framework Type	Representative Methods	Target Visibility	Box Prompt Type	Prediction Paradigm	Prompt Robustness
General Foundation	SAM [11], HQ-SAM [32]	Visible	Aligned, Precise	Direct	Low
General Prompt-Robust	PP-SAM [12], Stable-SAM [31]	Visible	Aligned, Noisy	Direct	Medium
Off-Nadir Task-Specific	OBM [13], PolyFootNet [26]	Occluded	Misaligned, Precise	Indirect	Medium
Off-Nadir Prompt-Robust	RBF-SAM (Ours)	Occluded	Aligned & Misaligned, Precise & Noisy	Indirect	High

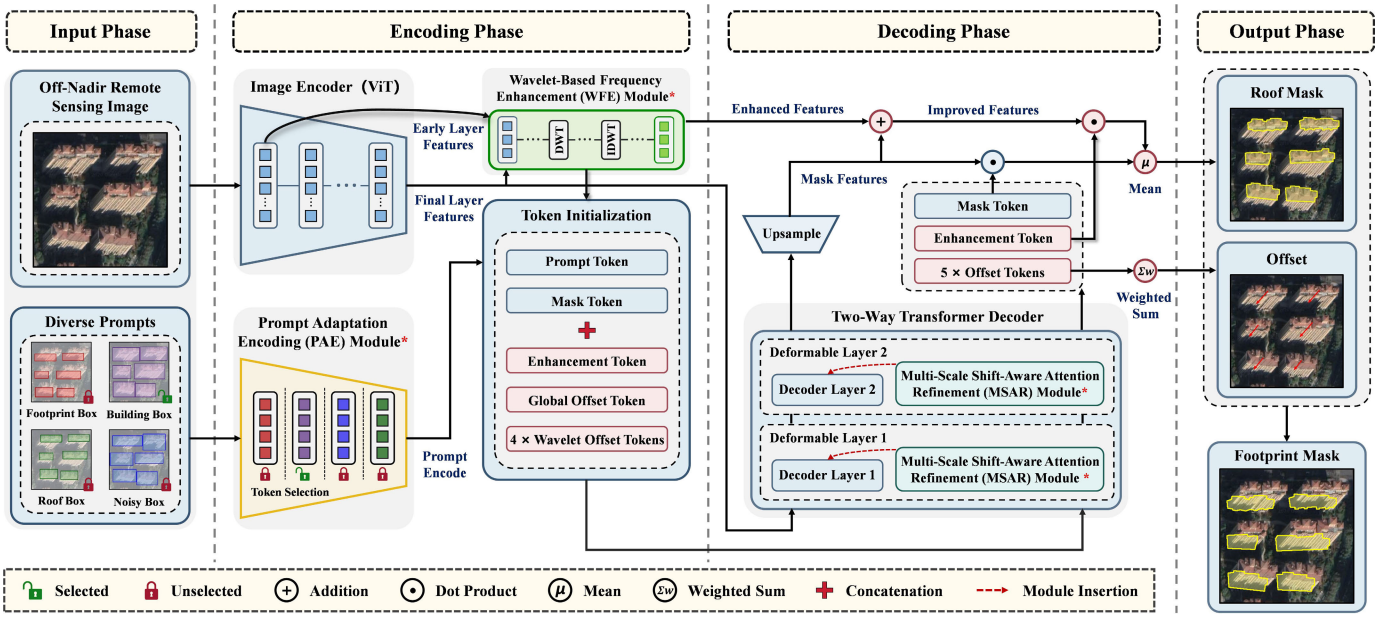


Fig. 2. Architecture of the proposed RBF-SAM. The standard SAM image encoder is used to extract high-level features, while the PAE module transforms diverse box-based prompts into learnable tokens. Within the decoder, the MSAR module is integrated to rectify attention shifts. Besides, the WFE module utilizes multifrequency features to enhance offset prediction and roof segmentation. Finally, the model is trained via the PRT strategy to ensure robustness to inaccurate prompts.

Table I further summarizes the fundamental differences among representative SAM-based frameworks.

To our knowledge, the robustness of SAM across varying prompt conditions in remote sensing, particularly under off-nadir scenarios, has not yet been comprehensively investigated. In this study, we explore a prompt-robust SAM framework for off-nadir building footprint extraction under diverse prompt qualities and types.

III. METHODOLOGY

A. Preliminary: SAM

As a prompt-driven foundation model for image segmentation, SAM [11] consists of three core components. 1) *Image Encoder:* A Vision Transformer (ViT) that is pretrained using the masked autoencoder (MAE) [55] approach to extract image features. 2) *Prompt Encoder:* An encoding module that processes interactive prompts, including both sparse types (i.e., points, boxes, and text) and dense types (i.e., masks). 3) *Mask Decoder:* A lightweight two-layer Transformer decoder that integrates image and prompt embeddings via a cross-attention mechanism and outputs a segmentation mask together with an intersection-over-union (IoU) score.

B. Overview of RBF-SAM

Following [17], we formulate building footprint extraction from off-nadir imagery as a joint task of instance-level roof segmentation and offset vector regression, while adhering to the original prompting paradigm of SAM. Fig. 2 illustrates the overall architecture of RBF-SAM, which enables robust and precise building segmentation through the coordinated optimization of prompt encoding, prompt–image interaction, and image representation. Specifically, given an off-nadir remote sensing image under diverse prompt conditions (i.e., bounding boxes of different types and qualities), RBF-SAM employs the standard SAM image encoder to extract image features. In parallel, the PAE module encodes input prompts into corresponding learnable tokens based on prompt categories. To achieve precise offset regression, we further introduce a zero-initialized global offset token and propose four frequency-directional wavelet offset tokens to capture specific directional details. Following this, the extracted image features, together with these five offset tokens and the enhancement token adapted from [32], are jointly fed into the decoder for inference. During the decoding phase, the MSAR module is integrated into the original cross-attention layers of the two-way Transformer decoder, enabling more accurate attention allocation and promoting feature aggregation on target

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regions. Additionally, the WFE module leverages multifrequency information to initialize the four frequency-directional wavelet offset tokens and to improve the feature representations after the upsampling stage. The improved features subsequently interact with the enhancement token to generate a high-quality mask, which is then averaged with the original mask to produce the refined roof segmentation. Meanwhile, the predictions derived from multiple offset tokens are aggregated via a weighted sum with learnable parameters to compute the final offset vector. Finally, the resulting building footprint is obtained by translating the roof mask according to the offset vector. Notably, based on [12] and [31], RBF-SAM is trained using a simple yet effective PRT strategy. Specifically, this strategy exposes the model to prompts randomly sampled from diverse types and qualities with equal probability during training, including various box-based forms such as ground-truth footprint, roof, and building boxes, as well as noisy bounding boxes, thereby enhancing the model's robustness to inaccurate prompts.

C. PAE Module

Given the significant diversity of input prompts in building footprint extraction from off-nadir remote sensing imagery, the unified bounding box encoding mechanism employed in the original SAM framework fails to effectively capture the intrinsic geometric and semantic discrepancies between distinct prompt types (e.g., roof versus footprint boxes). To overcome this limitation, the PAE module expands the prompting mechanism to support varying bounding box categories, while preserving SAM's original prompting paradigm. Specifically, it extends SAM's input protocol by associating each bounding box with a distinct category label (e.g., roof, footprint, or noisy prompts) during the input stage, thereby explicitly indicating its semantic properties. Based on this, the PAE module implements a category-aware conditional encoding strategy. Adhering to SAM's sparse embedding protocol, it encodes the top-left and bottom-right coordinates of the bounding box together with a category-specific learnable embedding corresponding to the box category label during both training and inference. Subsequently, these tokens are concatenated with other sparse prompts (e.g., point prompts) and jointly fed into the decoder to guide its attention to the prompt-specified region. By explicitly disentangling distinct bounding box types during the encoding phase, the PAE module enhances the model's adaptability to complex prompt conditions with negligible computational overhead, laying a robust foundation for subsequent attention refinement. Furthermore, to alleviate GPU memory constraints caused by dense bounding box prompts in remote sensing scenarios [25], we optimize the implementation of the SAM-based model for improved memory efficiency. Meanwhile, we adopt a uniform random sampling strategy during training, where a fixed number of prompts are sampled with equal probability from each image, ensuring prompt diversity while improving training efficiency.

D. MSAR Module

To mitigate attention drift caused by semantic inconsistencies arising from diverse prompts for the same target object, we introduce the MSAR module as a core component within the SAM decoder. Specifically, the MSAR module establishes a dynamic calibration mechanism between the original image features and those sampled via multiscale deformation, thereby ensuring the network focuses on the correct semantic regions in off-nadir imagery. As shown in Fig. 3, the MSAR module consists of two key parts: the multiscale deformable feature generator and the shift-aware attention refiner, which are described in detail below.

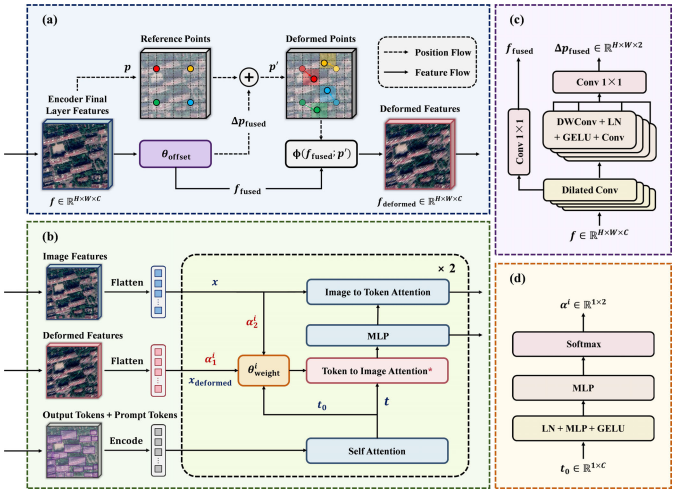


Fig. 3. Structure of the MSAR module. (a) Multiscale deformable feature generator, which adaptively samples encoder features to produce object-focused deformed features. (b) Shift-aware attention refiner, which integrates the deformed features into the decoder's cross-attention mechanism to rectify attention shifts. (c) and (d) Offset and weight prediction networks, respectively.

1) Multiscale Deformable Feature Generator: Specifically, given the output features $f \in \mathbb{R}^{H\times W\times C}$ from the SAM encoder, we first construct a uniform grid $p \in \mathbb{R}^{H\times W\times 2}$ to serve as spatial references. This grid comprises linearly spaced 2-D coordinates ranging from $(0,0)$ to $(H-1, W-1)$. We then normalize these coordinates to the $[-1,+1]$ range to facilitate subsequent sampling operations, thereby establishing a standard coordinate mapping where the top-left and bottom-right corners of the feature map correspond to $(-1,-1)$ and $(+1,+1)$, respectively. To generate offsets for each reference point within this coordinate space, we introduce an offset prediction network θ_{offset}, as illustrated in Fig. 3(c). Given the diverse scales of buildings in remote sensing imagery, this network employs a multiscale dilated convolution strategy, feeding the original features f into parallel branches to extract distinct scale-aware contexts f_n. These contexts are subsequently mapped into multiscale offsets through a simple depthwise convolutional block. To account for the varying offset ranges captured by different branches, we utilize a hyperbolic tangent function (tanh) modulated by predefined scaling factors s_n to dynamically constrain the output amplitudes, yielding the constrained offsets Δp_n

$$\Delta p_n = s_n \cdot \tanh\left(\theta_{\text{offset}}(f)_n\right) \tag{1}$$

where n denotes the nth branch. This mechanism effectively prevents sampling out-of-bounds while ensuring stability during training. Finally, we employ 1×1 convolutional layers

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to independently aggregate the multibranch feature maps f_n and predicted offsets Δp_n. This generates the global fused features f_{fused} and the final fused offset field Δp_{fused}, which are formulated as

$$f_{\text{fused}} = \text{Conv}_{1\times1}\left(\text{Concat}\left(f_1, f_2, \ldots, f_N\right)\right) \tag{2}$$

$$\Delta p_{\text{fused}} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\Delta p_1, \Delta p_2, \ldots, \Delta p_N\right)\right) \tag{3}$$

where Concat$(\cdot)$ denotes the concatenation operation along the channel dimension and N represents the total number of parallel branches. Consequently, we apply the fused offset field Δp_{fused} to the original reference grid p, resulting in the updated sampling locations $p' = p + \Delta p_{\text{fused}}$. To accommodate the noninteger coordinates of p', we follow [56] and employ a bilinear interpolation $\Phi(\cdot;\cdot)$ to ensure the differentiable resampling of the fused features f_{fused}. Specifically, for any updated position (p'_x, p'_y), the output features f_{deformed} are computed as a weighted average of its four nearest integer points from the fused features f_{fused}

$$\begin{aligned} f_{\text{deformed}} &= \Phi\left(f_{\text{fused}}; p'\right) \\ &= \sum_{(r_x, r_y)} g\left(p'_x, r_x\right) g\left(p'_y, r_y\right) f_{\text{fused}}\left[r_y, r_x, :\right] \end{aligned} \tag{4}$$

where $g(a,b) = \max(0, 1 - |a - b|)$ denotes the interpolation kernel function and (r_x, r_y) indexes all the locations on the fused features f_{fused}. Through this process, we ultimately obtain deformed features f_{deformed} that adaptively focus on target objects, providing target-enriched context for the subsequent shift-aware attention refiner.

2) Shift-Aware Attention Refiner: We utilize the extracted deformed features to update the original token-to-image cross-attention module within SAM. This module employs prompt tokens as queries to attend to image features, aggregating relevant visual information. Specifically, given the prompt token features $t \in \mathbb{R}^{T\times C}$ and the flattened image features $x \in \mathbb{R}^{HW\times C}$, the original cross-attention mechanism can be formulated as

$$\text{CAttn}\left(t, x\right) = \sigma\left(\frac{Q\left(t\right)K\left(x\right)^T}{\sqrt{d}}\right)V\left(x\right) \tag{5}$$

where $\sigma(\cdot)$ represents the Softmax function, d is the projected channel dimension, and $Q(\cdot), K(\cdot)$, and $V(\cdot)$ denote the linear projection functions for queries, keys, and values, respectively.

As observed in (5), the keys and values in the original design depend exclusively on x. Our objective is to integrate the original features x with the flattened deformable features $x_{\text{deformed}} \in \mathbb{R}^{HW\times C}$ to correct attention shifts. Therefore, to independently adapt the keys and values, we employ two parallel lightweight networks with identical architectures, denoted as θ^k_{weight} and θ^v_{weight}, as detailed in Fig. 3(d). These predictors process the prompt token $t_0 \in t$ through a feed-forward network (FFN) to predict adaptive coefficients, yielding the final weights via Softmax

$$\alpha^i = \sigma\left(\theta^i_{\text{weight}}\left(t_0\right)\right), \quad i \in \{k, v\} \tag{6}$$

where $\alpha^i = [\alpha^i_1, \alpha^i_2] \in \mathbb{R}^{1\times2}$ denotes the predicted weighting coefficients for the keys $(i = k)$ and values $(i = v)$. Subsequently, based on the predicted α^k and α^v, we fuse the original and deformable features directly within the attention mechanism. The updated token-to-image cross-attention mechanism unfolds as follows:

$$\tilde{x}^k = \alpha^k_1 x_{\text{deformed}} + \alpha^k_2 x, \quad \tilde{x}^v = \alpha^v_1 x_{\text{deformed}} + \alpha^v_2 x \tag{7}$$

$$\text{CAttn}^*\left(t, x, x_{\text{deformed}}\right) = \sigma\left(\frac{Q\left(t\right)K\left(\tilde{x}^k\right)^T}{\sqrt{d}}\right)V\left(\tilde{x}^v\right). \tag{8}$$

Through the above design, the MSAR module enables RBF-SAM to selectively leverage adaptive sampling offsets to recalibrate the attention mechanism, improving the robustness of building footprint segmentation in off-nadir imagery while preserving SAM's inherent zero-shot generalization capability.

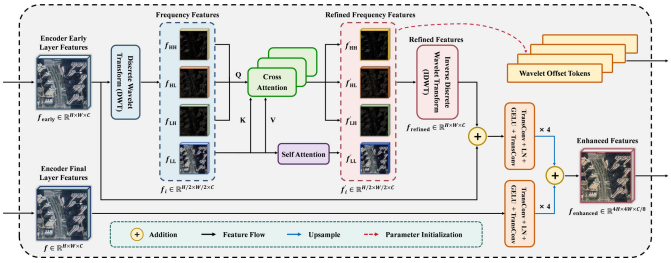


Fig. 4. Structure of the WFE module. By leveraging wavelet-based frequency decomposition, interaction, and reconstruction, the module captures rich structural and detail information. The resulting features are further used to initialize the wavelet offset tokens and reinforce the upsampled feature representations.

E. WFE Module

To effectively exploit frequency information for refining offset prediction and enhancing roof segmentation, we propose the WFE module. As illustrated in Fig. 4, we extract early layer features $f_{\text{early}} \in \mathbb{R}^{H\times W\times C}$ from the SAM encoder after the first global attention block, as they preserve more fine-grained spatial details [57]. Subsequently, we perform the discrete wavelet transform (DWT) on these features for frequency decomposition [58], [59]

$$f_{\text{LL}}, f_{\text{LH}}, f_{\text{HL}}, f_{\text{HH}} = \text{DWT}\left(f_{\text{early}}\right) \tag{9}$$

where f_{LL} corresponds to the low-frequency component, whereas f_{LH}, f_{HL}, and f_{HH} denote the horizontal, vertical, and diagonal high-frequency details, respectively.

Recognizing that low-frequency features capture global context while high-frequency components correspond to local textures such as edges, we design a global–local interactive attention mechanism, inspired by [49] and [60]. Specifically, we first apply self-attention to f_{LL} to capture global dependencies, yielding the refined feature f'_{LL}, computed as

$$f'_{\text{LL}} = \sigma\left(\frac{Q\left(f_{\text{LL}}\right)K\left(f_{\text{LL}}\right)^T}{\sqrt{d}}\right)V\left(f_{\text{LL}}\right). \tag{10}$$

Meanwhile, we formulate a cross-attention mechanism where the high-frequency detail components individually act as queries, while the low-frequency f_{LL} serves as keys and values to provide matching global contexts, yielding the corresponding refined components f'_{LH}, f'_{HL}, and f'_{HH}

$$f'_i = \sigma\left(\frac{Q\left(f_i\right)K\left(f_{\text{LL}}\right)^T}{\sqrt{d}}\right)V\left(f_{\text{LL}}\right), \quad i \in \{\text{LH}, \text{HL}, \text{HH}\}. \tag{11}$$

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Following the attention-based interactions, the refined components are reconstructed back to the spatial domain via the inverse DWT (IDWT), yielding the refined spatial features f_{refined}

$$f_{\text{refined}} = \text{IDWT}\left(f'_{\text{LL}}, f'_{\text{LH}}, f'_{\text{HL}}, f'_{\text{HH}}\right). \tag{12}$$

The refined features derived above serve a dual purpose within our framework. On the one hand, the four refined frequency components $f'_{\text{LL}}, f'_{\text{LH}}, f'_{\text{HL}}, f'_{\text{HH}}$ are individually utilized to initialize four wavelet offset tokens. Subsequently, together with a spatial-level global offset token, these tokens generate parallel offset predictions, which are then fused through a learnable weighted sum. On the other hand, f_{refined} and the encoder output f are fused through simple transposed convolutional layers to yield f_{enhanced}. To alleviate spatial detail degradation during upsampling, f_{enhanced} is further combined with the original upsampled features via element-wise summation. Finally, following [32], the resulting features interact with a newly introduced enhancement token in the mask prediction layer to produce high-quality masks. Furthermore, since maintaining output ambiguity is unnecessary for this task [13], [26], we directly use the first predicted mask as the final prediction.

F. Loss Function

We formulate the optimization of RBF-SAM as an end-to-end multitask learning problem. The overall loss function, denoted as $\mathcal{L}_{\text{total}}$, is defined as a weighted linear combination of the segmentation loss $\mathcal{L}_{\text{seg}}$ for roof extraction and the regression loss $\mathcal{L}_{\text{offset}}$ for offset learning

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{seg}} + \lambda_2 \mathcal{L}_{\text{offset}}. \tag{13}$$

For roof segmentation, following [32], we employ a combination of binary cross-entropy (BCE) [61] and Dice [62] losses to ensure both pixel-level accuracy and region-level alignment

$$\mathcal{L}_{\text{seg}} = \lambda_3 \mathcal{L}_{\text{bce}} + \lambda_4 \mathcal{L}_{\text{dice}}. \tag{14}$$

Regarding offset regression, the Smooth L1 loss [63] is utilized. Specifically, the loss is composed of supervision on the offset predictions from different wavelet tokens $\mathcal{L}^i_{\text{wave}}$ (where $i = 1, \ldots, N$ corresponds to each direction and $N = 4$), the global offset token $\mathcal{L}_{\text{global}}$, and the final refined offset $\mathcal{L}_{\text{final}}$. This is formulated as

$$\mathcal{L}_{\text{offset}} = \lambda_5 \sum_{i=1}^{N} \mathcal{L}^i_{\text{wave}} + \lambda_6 \mathcal{L}_{\text{global}} + \lambda_7 \mathcal{L}_{\text{final}}. \tag{15}$$

In our experiments, the loss weights λ_i are configured to ensure balanced optimization. Specifically, we follow [17] to set the task-level weights to $\lambda_1 = 1$ and $\lambda_2 = 16$. The remaining weights are empirically determined: we assign $\lambda_3 = 0.5$ and $\lambda_4 = 10$ for segmentation to balance pixel-level and region-level supervision, while setting $\lambda_5 = \lambda_6 = 1$ and $\lambda_7 = 2$ for offset regression to prioritize the final output.

IV. Experiments and Analysis

A. Dataset

To evaluate the performance and generalization capability of the proposed method, extensive experiments are conducted on three publicly available benchmark datasets: the BONAI [17] dataset, the OmniCity-view3 [64] dataset, and the Huizhou [13] test set. Further details are provided below.

1) BONAI Dataset: This dataset [17] is the first benchmark dedicated to building footprint extraction from off-nadir remote sensing images. It comprises 3300 images of size 1024×1024 pixels, containing approximately 270 000 annotated building instances. The 3000 training and validation images cover five representative Chinese cities: Beijing, Shanghai, Harbin, Jinan, and Chengdu, while the test set includes 300 images from Shanghai and Xi'an. The dataset provides instance-level annotations, including roof segmentation, footprint segmentation, offset vectors, and building height.

2) OmniCity-view3 Dataset: Originally collected from New York City, this dataset [64] consists of satellite images of size 512×512 pixels. In this study, we utilize the extended version from [18], which incorporates additional offset annotations and comprises 17 092 images for training and validation, along with 4929 images for testing. Available annotations cover roof and footprint segmentation, building height, and roof-to-footprint offsets.

3) Huizhou Test Set: Similar to BONAI, this dataset [13] includes manual annotations for roof and footprint segmentation, along with their corresponding offset vectors. It comprises images of size 1024×1024 pixels collected from Huizhou, China, and contains over 7000 building instances.

B. Evaluation Metrics

Consistent with [13] and [26], we evaluate the effectiveness of RBF-SAM and comparative methods across two dimensions: footprint extraction quality and offset estimation accuracy. For footprint quality, we employ precision, recall, and $F1$-score, with the $F1$-score serving as the primary metric [17]. To quantify offset estimation accuracy, we calculate the vector error (VE), length error (LE), and angle error (AE), defined as follows:

$$\text{VE} = |\vec{p} - \vec{p}_g|_2; \quad \text{LE} = ||\vec{p}|_2 - |\vec{p}_g|_2|; \quad \text{AE} = |\theta - \theta_g| \tag{16}$$

where $\vec{p}$ and θ denote the predicted offset vector and angle, respectively, while $\vec{p}_g$ and θ_g represent their corresponding ground-truth values. The symbols $|\cdot|$ and $|\cdot|_2$ denote the L_1-norm and the L_2-norm, respectively. Following the COCO metric [65], we evaluate offsets across different length intervals and derive an average score

$$m\mathcal{E} = \frac{\mathcal{E}_{(10n,\infty)} + \sum_{i=0}^{n} \mathcal{E}_{(10i,10i+10)}}{n+1} \tag{17}$$

where $\mathcal{E}$ represents VE, LE, or AE. The term $\mathcal{E}_{(i,j)}$ represents the mean error for the offset group with lengths ranging from i to j, partitioned with a step size of 10. Specifically, to quantify the overall deviation, we compute $\mathcal{E}_{(0,\infty)}$, which corresponds to the overall average error, denoted as $a\mathcal{E}$. Note that these offset-based metrics require a one-to-one correspondence between predictions and ground-truth instances [30], and are thus reported only for promptable models.

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C. Experimental Settings

We conduct experiments using the PyTorch [66] framework on four NVIDIA L20 GPUs. The model is trained for 48 epochs with a batch size of 4. We employ stochastic gradient descent (SGD) [67] as the optimizer, with a momentum of 0.9 and a weight decay of 0.0001. The learning rate is initialized at 0.0025 and is decayed by a factor of 0.1 at the 32nd and 44th epochs, following a 500-iteration warmup. For data augmentation, we apply random horizontal and vertical flips.

To align with the standard input size of the SAM-based model, all images are resized to 1024×1024 pixels. For both prompt robustness analysis and ablation studies, models are conditioned on diverse box-based prompts: ground-truth footprint, roof, and building boxes, as well as noisy boxes. To simulate practical annotation uncertainty in remote sensing, motivated by the experimental design in [13], noisy boxes are generated by randomly perturbing the boundaries of the ground-truth building box within a range of $[-15, +15]$ pixels. By default, we adopt the ground-truth footprint prompt as the optimal prompt setting for RBF-SAM based on the $F1$-score.

D. Comparison Models

To comprehensively validate the effectiveness of the proposed RBF-SAM, we conduct comparative experiments with a wide range of state-of-the-art methods, which can be categorized into two groups.

1) Automatic Instance Segmentation Methods: It encompasses both generic instance segmentation models, including Mask R-CNN [46], PANet [68], and HRNet [69], and methods specifically designed for building footprint extraction from off-nadir images, such as LOFT [17], CasLOFT [13], and MLS-BRN [18].

2) Promptable Instance Segmentation Methods: This category comprises foundation models represented by SAM [11] together with its variants HQ-SAM [32] and PP-SAM [12], all implemented with the ViT-Base backbone to align with our backbone setting, alongside prompt-driven methods tailored for remote sensing tasks, including prompt LOFT (p.LOFT), prompt cascade LOFT (p.CasLOFT), and the SAM-based frameworks OBM [13] as well as PolyFootNet [26].

To ensure fair comparison, we evaluate the performance of all methods within a unified PyTorch environment on the MMDetection [70] platform. For general instance segmentation methods, the models are trained using default settings. Regarding domain-specific algorithms in remote sensing, we either directly evaluate the officially released model weights or reproduce the models using their open-source codebases, thereby ensuring the comparability of experimental results.

E. Performance Evaluation

1) Comparison With State-of-the-Art Models: We compare the proposed RBF-SAM with state-of-the-art methods on the BONAI and OmniCity-view3 datasets, where each promptable method is evaluated using its optimal prompt setting in terms of the $F1$-score. As shown in Table II, RBF-SAM achieves the highest $F1$-scores on both datasets, indicating its effectiveness in delineating building footprints under challenging off-nadir conditions. Specifically, compared with the state-of-the-art method on the BONAI dataset, RBF-SAM improves the $F1$-score, precision, and recall by 4.57%, 3.21%, and 5.81%, respectively. On the OmniCity-view3 dataset, it achieves corresponding improvements of 1.89%, 1.97%, and 1.80%. These gains benefit from the WFE module in RBF-SAM, which integrates multifrequency information to facilitate more accurate building footprint delineation. Furthermore, the results indicate that leveraging SAM-based foundation models or integrating an offset regression head generally leads to improved performance compared with traditional direct footprint prediction approaches. Table III compares the offset prediction accuracy of RBF-SAM with other promptable approaches that incorporate offset prediction. On the BONAI dataset, RBF-SAM achieves performance comparable to state-of-the-art methods, with an aVE difference of only 0.01 pixels compared to OBM and a reduction of 0.37 pixels relative to PolyFootNet. It also yields near-best performance in aLE and aAE, with differences of at most 0.02. On the OmniCity-view3 dataset, RBF-SAM outperforms all competing models

TABLE II

QUANTITATIVE COMPARISON OF BUILDING FOOTPRINT EXTRACTION PERFORMANCE WITH STATE-OF-THE-ART METHODS ON THE BONAI AND OMNICITY-VIEW3 DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	BONAI dataset			OmniCity-view3 dataset		
	F1-score	Precision	Recall	F1-score	Precision	Recall
I. Automatic methods without offset prediction						
Mask R-CNN [46]	62.88	67.84	60.98	71.72	75.79	70.82
PANet [68]	62.95	67.99	60.99	71.19	76.25	69.42
HRNetv2 [69]	62.67	68.51	60.40	73.08	77.49	71.91
II. Automatic methods with offset prediction						
LOFT [17]	65.60	69.05	64.87	73.80	76.25	73.87
CasLOFT [13]	59.12	70.47	54.02	69.02	79.34	63.65
MLS-BRN [18]	65.91	69.83	65.01	76.23	73.82	81.62
III. Promptable methods without offset prediction						
SAM [11]	75.82	71.03	82.73	78.25	82.62	74.81
HQ-SAM [32]	76.03	71.17	83.05	80.45	85.31	76.52
PP-SAM [12]	80.16	**90.04**	72.39	82.75	90.58	76.40
IV. Promptable methods with offset prediction						
p.LOFT [13]	73.98	85.66	65.64	82.15	90.70	75.57
p.CasLOFT [13]	77.34	87.42	69.76	83.09	91.11	76.83
OBM [13]	81.26	82.43	80.37	86.05	90.17	82.53
PolyFootNet [26]	78.74	79.85	77.91	88.42	90.06	87.01
RBF-SAM (ours)	**85.83**	85.64	**86.18**	**90.31**	**92.03**	**88.81**

TABLE III

QUANTITATIVE COMPARISON OF OFFSET PREDICTION PERFORMANCE WITH STATE-OF-THE-ART METHODS ON THE BONAI AND OMNICITY-VIEW3 DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	BONAI dataset						OmniCity-view3 dataset					
	aVE	aLE	aAE	mVE	mLE	mAE	aVE	aLE	aAE	mVE	mLE	mAE
p.LOFT [13]	6.22	4.65	0.31	15.62	12.75	0.18	7.82	5.26	0.71	51.75	**42.99**	0.65
p.CasLOFT [13]	5.84	4.38	0.30	**14.61**	**12.28**	0.17	7.82	5.42	0.69	**51.06**	43.37	0.61
OBM [13]	**5.08**	**4.03**	**0.22**	14.85	13.55	0.12	6.69	5.15	0.64	56.88	53.67	0.66
PolyFootNet [26]	5.46	4.40	**0.22**	17.0	15.8	**0.11**	6.15	4.65	0.59	51.5	47.9	0.63
RBF-SAM (ours)	5.09	4.05	0.23	15.11	13.75	0.12	**5.84**	**4.55**	**0.48**	53.59	50.66	**0.57**

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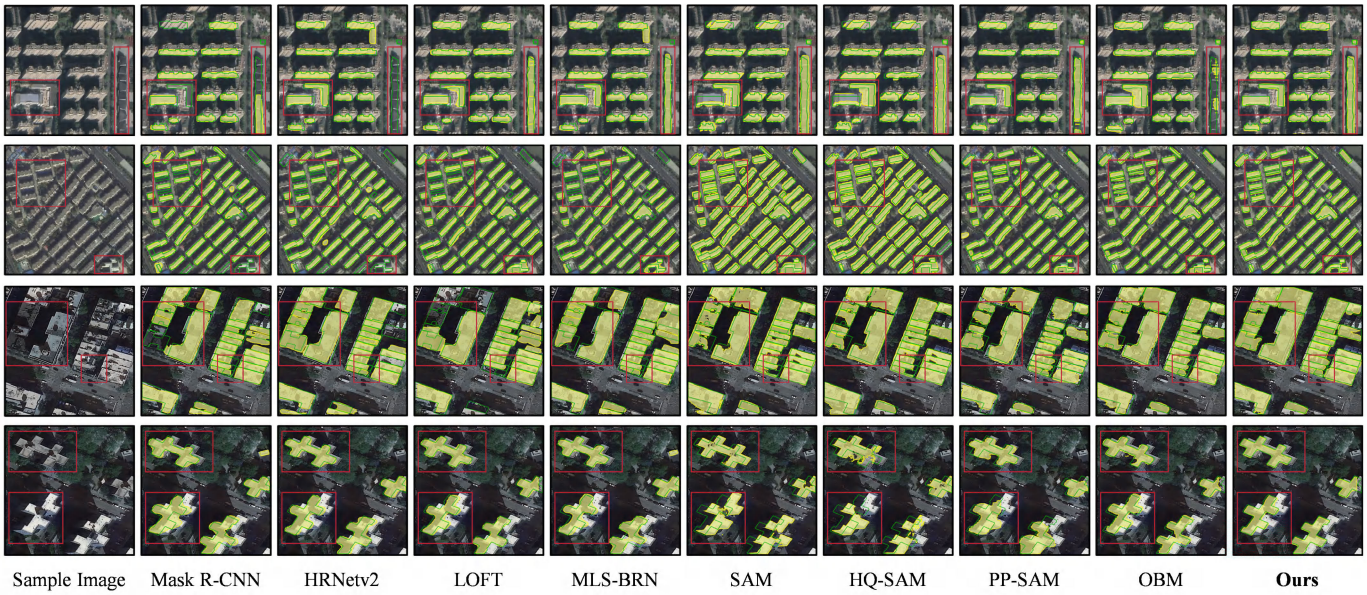


Fig. 5. (First two rows) Comparison of building footprint extraction results between the proposed RBF-SAM and other state-of-the-art methods on the BONAI and (last two rows) OmniCity-view3 datasets. The yellow and green colors represent the predicted results and the ground truth, respectively.

TABLE IV

ROBUSTNESS COMPARISON OF BUILDING FOOTPRINT EXTRACTION WITH PROMPTABLE METHODS ON THE BONAI AND OMNICITY-VIEW3 DATASETS ACROSS DIVERSE PROMPT SETTINGS. THE BEST RESULTS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	Method	Footprint box			Roof box			Building box			Noisy box		
		F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall
BONAI	SAM [11]	75.82	71.03	82.73	63.68	59.54	69.34	70.65	60.58	86.56	68.53	59.56	82.46
	HQ-SAM [32]	76.03	71.17	83.05	64.02	59.99	69.37	70.97	60.61	**87.33**	68.85	59.62	**83.13**
	PP-SAM [12]	80.16	90.04	72.39	60.88	69.46	54.33	77.24	78.17	77.30	70.53	72.59	69.52
	p.LOFT [13]	64.51	88.84	51.78	68.82	**83.00**	59.19	73.98	85.66	65.64	70.32	**85.08**	60.44
	p.CasLOFT [13]	66.38	**90.68**	53.37	65.95	79.16	56.79	77.34	**87.42**	69.76	71.28	84.04	62.27
	OBM [13]	72.80	84.37	64.68	66.02	73.06	60.42	81.26	82.43	80.37	72.30	75.01	70.05
	RBF-SAM (ours)	**85.83**	85.64	**86.18**	**80.88**	80.31	**81.59**	**83.71**	83.37	84.29	**79.52**	79.66	79.63
OmniCity-view3	SAM [11]	77.60	85.74	71.39	75.73	82.75	70.20	78.25	82.62	74.81	77.78	82.61	73.99
	HQ-SAM [32]	80.45	85.31	76.52	78.00	82.35	74.43	80.43	82.14	79.25	80.02	82.14	78.47
	PP-SAM [12]	82.75	90.58	76.40	77.36	85.53	70.83	82.75	86.43	79.75	81.35	86.05	77.56
	p.LOFT [13]	81.47	91.68	73.78	81.72	90.81	74.76	82.15	90.70	75.57	81.69	90.70	74.80
	p.CasLOFT [13]	82.04	92.24	74.34	82.02	91.04	75.07	83.09	91.11	76.83	82.35	91.08	75.60
	OBM [13]	83.73	**92.41**	76.84	81.25	89.70	74.52	86.05	90.17	82.53	84.23	89.86	79.55
	RBF-SAM (ours)	**90.31**	92.03	**88.81**	**89.19**	**91.07**	**87.57**	**89.75**	**91.67**	**88.10**	**89.34**	**91.38**	**87.58**

across aVE, aLE, aAE, and mAE metrics, while remaining competitive on mVE and mLE. Specifically, it reduces the aVE by 0.31 pixels and 0.85 pixels compared with PolyFootNet and OBM, respectively. It is worth noting that results for RBF-SAM are reported under the optimal prompt setting for fair comparison. Using ground-truth building box prompts further improves offset accuracy, reducing the lowest aVE on the BONAI dataset by 0.03 pixels. Detailed analysis is provided in Section IV-E2.

To provide a comprehensive qualitative analysis, Fig. 5 visualizes the building extraction results of RBF-SAM compared to several state-of-the-art methods on the BONAI and OmniCity-view3 datasets. Overall, our proposed method produces highly accurate predictions, yielding more complete building delineations with a notably reduced false negative rate. Specifically, on the BONAI dataset, RBF-SAM demonstrates strong robustness to shadow occlusions in high-rise building areas (the first row) and effectively handles the extraction challenges in dense building distributions (the second row). Furthermore, on the higher-resolution OmniCity-view3 dataset, while existing methods often suffer from omission errors due to complex backgrounds, RBF-SAM accurately identifies and delineates individual buildings (the third row). Moreover, it effectively captures the fine-grained footprint contours of complex structures (the fourth row).

2) Analysis of Prompt Robustness: To systematically evaluate the robustness of RBF-SAM under different prompting conditions, we conduct comparative experiments on the BONAI and OmniCity-view3 datasets using various types of box prompts as inputs, including ground-truth footprint, roof, and building boxes, as well as noisy boxes. Table IV reports the footprint segmentation performance, while Table V presents the corresponding offset prediction errors. The experimental results demonstrate that RBF-SAM maintains a stable performance advantage even when existing methods are provided with high-quality prompts consistent with their training

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TABLE V

ROBUSTNESS COMPARISON OF OFFSET PREDICTION WITH PROMPTABLE METHODS ON THE BONAI AND OMNICITY-VIEW3 DATASETS ACROSS DIVERSE PROMPT SETTINGS. THE BEST RESULTS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	Method	Footprint box			Roof box			Building box			Noisy box		
		aVE	aLE	aAE	aVE	aLE	aAE	aVE	aLE	aAE	aVE	aLE	aAE
BONAI	p.LOFT [13]	7.26	5.61	0.33	7.73	6.18	0.33	6.22	4.65	0.31	6.42	4.78	0.33
	p.CasLOFT [13]	7.69	6.17	0.34	8.68	7.24	0.35	5.84	4.38	0.30	6.54	4.86	0.34
	OBM [13]	7.45	6.58	0.24	9.13	8.29	0.25	5.08	4.03	**0.22**	5.93	4.75	**0.24**
	RBF-SAM (ours)	**5.09**	**4.05**	**0.23**	**5.58**	**4.48**	**0.24**	**5.05**	**3.96**	0.23	**5.37**	**4.25**	**0.24**
OmniCity-view3	p.LOFT [13]	7.72	5.34	0.71	7.90	5.51	0.72	7.82	5.26	0.71	7.84	5.27	0.71
	p.CasLOFT [13]	7.66	5.42	0.68	8.04	5.67	0.72	7.82	5.42	0.69	7.86	5.44	0.70
	OBM [13]	6.59	5.12	0.63	7.00	5.60	0.65	6.69	5.15	0.64	6.74	5.21	0.64
	RBF-SAM (ours)	**5.84**	**4.55**	**0.48**	**6.07**	**4.72**	**0.50**	**5.88**	**4.56**	**0.50**	**5.91**	**4.59**	**0.50**

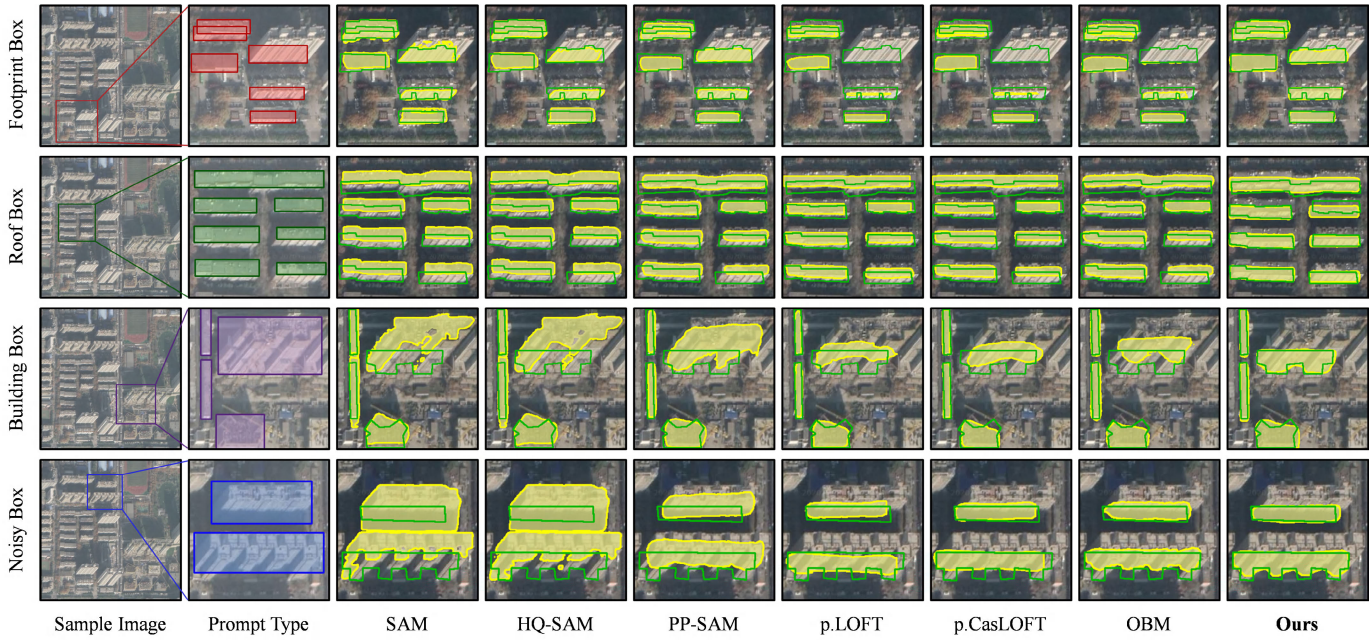


Fig. 6. Robustness evaluation of the proposed RBF-SAM and other promptable methods under varying prompt conditions on the BONAI dataset, where the predicted results and the ground truth are highlighted in yellow and green, respectively.

stage (i.e., ground-truth building boxes). Compared with the second-best method OBM, RBF-SAM achieves $F1$-score improvements of 2.45% on the BONAI dataset and 3.70% on the OmniCity-view3 dataset. However, as the prompt type changes or the prompt quality degrades, the prediction performance of existing methods deteriorates significantly. For example, on the BONAI dataset, when OBM utilizes roof boxes as prompts, its $F1$-score drops sharply from 81.26% to 66.02%, corresponding to a performance degradation of 15.24%. In contrast, RBF-SAM maintains stable segmentation performance and offset prediction accuracy across diverse prompt types, demonstrating superior prompt robustness. Specifically, compared with OBM, RBF-SAM achieves $F1$-score improvements of 13.03%, 14.86%, and 7.22% on the BONAI dataset under footprint box, roof box, and noisy box prompts, respectively. On the OmniCity-view3 dataset, RBF-SAM also yields increases of 6.58%, 7.94%, and 5.11% under footprint box, roof box, and noisy box prompts, respectively. Moreover, as shown in Table V, RBF-SAM attains the best or near-best performance across all offset-related metrics on both datasets under diverse prompt settings, further confirming its robustness to prompt variations.

Fig. 6 presents a qualitative comparison of building footprint extraction results from state-of-the-art methods under various prompt conditions. As observed, RBF-SAM consistently delineates building footprints with complete structures and coherent boundaries across diverse prompt settings, significantly outperforming existing methods. This robustness underscores the practical value of RBF-SAM in real-world remote sensing applications, where ground-truth annotations inevitably suffer from spatial inaccuracies or category inconsistencies. Moreover, we observe an interesting phenomenon: footprint box prompts yield the best performance, whereas roof box prompts result in the worst, despite being semantically more precise. This behavior likely stems from the model inference mechanism, where predictions rely on both spatial constraints and offset correction, and different prompts modulate the reliance on offsets. A detailed discussion is provided in Section V-E.

3) Analysis of Generalization and Complexity: To evaluate model generalization, models trained on the BONAI dataset are directly applied to the Huizhou test set without any additional fine-tuning, using the optimal prompt setting for each method. Table VI summarizes the footprint segmentation accuracy and model complexity. As shown, RBF-SAM achieves the

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Fig. 7. Generalization performance of the proposed RBF-SAM and other promptable methods on the Huizhou test set, where the predicted results and the ground truth are highlighted in yellow and green, respectively.

TABLE VI

GENERALIZATION PERFORMANCE COMPARISON WITH PROMPTABLE METHODS ON THE HUIZHOU TEST SET. MODEL COMPLEXITY IS REPORTED IN TERMS OF PARAMETER COUNT AND FPS, MEASURED USING A SINGLE NVIDIA L20 GPU. THE BEST RESULTS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	Params (M)	FPS	F1-score	Precision	Recall
SAM [11]	90.49	6.0	79.09	75.64	83.09
HQ-SAM [32]	91.56	3.8	78.53	75.69	81.96
PP-SAM [12]	90.49	6.0	78.04	**87.97**	70.31
p.LOFT [13]	**81.25**	**8.7**	71.04	82.38	62.87
p.CasLOFT [13]	189.30	7.2	73.85	82.92	67.14
OBM [13]	94.79	5.4	81.50	78.41	85.20
PolyFootNet [26]	115.20	-	80.36	78.99	82.18
RBF-SAM (ours)	101.46	3.9	**85.33**	83.38	**87.55**

best overall performance on this unseen dataset, surpassing the second-best performing method by 3.83%, 4.97%, and 2.35% in the $F1$-score, precision, and recall, respectively. These results indicate that RBF-SAM generalizes effectively across different urban scenes and imaging conditions, without relying on dataset-specific adaptation. In terms of model complexity, RBF-SAM maintains a moderate parameter count of 101.46 M with an inference speed of 3.9 FPS. Specifically, the PAE module introduces negligible computational cost, while the MSAR and WFE modules incur inference overheads of 0.7 and 1.2 FPS, respectively. Despite this additional computational cost, RBF-SAM remains slightly faster than HQ-SAM (3.8 FPS), demonstrating a reasonable balance between extraction accuracy and computational efficiency. Furthermore, its speed remains competitive with representative automatic methods such as Mask R-CNN (3.5 FPS) and LOFT (2.4 FPS), making it suitable for annotation workflows [71] and remote sensing interpretation tasks [72]. Qualitative results of building footprint extraction under the generalization setting are presented in Fig. 7.

V. DISCUSSION

A. Influence of Individual Components

We conduct a series of ablation studies under the PRT strategy to validate the effectiveness of each component of RBF-SAM. The ablation results for segmentation performance are presented in Table VII, while the offset prediction results are detailed in Table VIII. Following [13], we establish a baseline model by incorporating offset tokens, along with the original prompt tokens and mask tokens, into the SAM decoder for offset prediction. Experimental results demonstrate that, compared to the baseline, the introduction of the PAE module yields significantly improved robustness across diverse prompt conditions. Specifically, benefiting from the decoupled representations enabled by distinct learnable prompt tokens, $F1$-scores increase by 2.08%, 0.50%, 0.60%, and 0.19% under footprint box, roof box, building box, and noisy box prompts, respectively. Further incorporating the MSAR module leads to additional performance gains, with $F1$-score improvements of 1.54%, 0.79%, 1.65%, and 1.27% under footprint box, roof box, building box, and noisy box prompts, respectively, while the corresponding aVE reductions are 0.50, 0.54, 0.52, and 0.41. These results indicate that the proposed dynamic calibration mechanism effectively guides the network to focus on the true target regions under inaccurate prompts. Additionally, the inclusion of the WFE module yields further $F1$-score improvements ranging from 0.90% to 1.39%, accompanied by aVE reductions between 0.33 and 0.47 across different prompt conditions. These results confirm that integrating multifrequency information via wavelet transformation enhances boundary delineation and geometric correction. Fig. 8 visualizes the qualitative results under different ablation configurations, where the predictions and ground truth are highlighted in yellow and green, respectively.

B. Influence of MSAR Module Parameters

Table IX summarizes the segmentation and offset prediction results on the BONAI dataset under different MSAR module parameter configurations, with a focused analysis of dilation rates and offset factors. For the dilation rates, we compare three settings: $[1, 1, 1, 1]$, $[1, 2, 3, 4]$, and $[1, 3, 5, 7]$. Among them, the $[1, 2, 3, 4]$ configuration achieves the best performance by balancing local details and global context via progressively increasing rates. In contrast, the $[1, 1, 1, 1]$ setting is constrained by a limited receptive field, while the $[1, 3, 5, 7]$ configuration is more susceptible to interference from neighboring buildings. For the offset factors, we further evaluate three configurations: $[1, 1, 1, 1]$, $[0.5, 1, 2, 4]$, and $[1, 2, 4, 8]$. Compared to the fixed setting $[1, 1, 1, 1]$, the progressive offset factors in $[0.5, 1, 2, 4]$ better adapt to building structures of varying scales and shapes. Conversely, the large span in $[1, 2, 4, 8]$ leads to an overly broad sampling range, introducing redundant information. Consequently, we select the $[1, 2, 3, 4]$ configuration for dilation rates and the $[0.5, 1, 2, 4]$ setting for offset factors as the default setup for the MSAR module.

C. Influence of the PRT Strategy

Robust training is pivotal for segmentation stability under varying prompts, yet often overlooked in prior studies. Existing methods typically rely on high-quality prompts consistent with the training phase, leading to significant performance degradation when inputs exhibit spatial inaccuracies or category inconsistencies [12], [31]. Fig. 9 presents a comparative analysis of RBF-SAM with and without the PRT strategy in terms of segmentation performance and offset prediction accuracy, where results are averaged over five random seeds,

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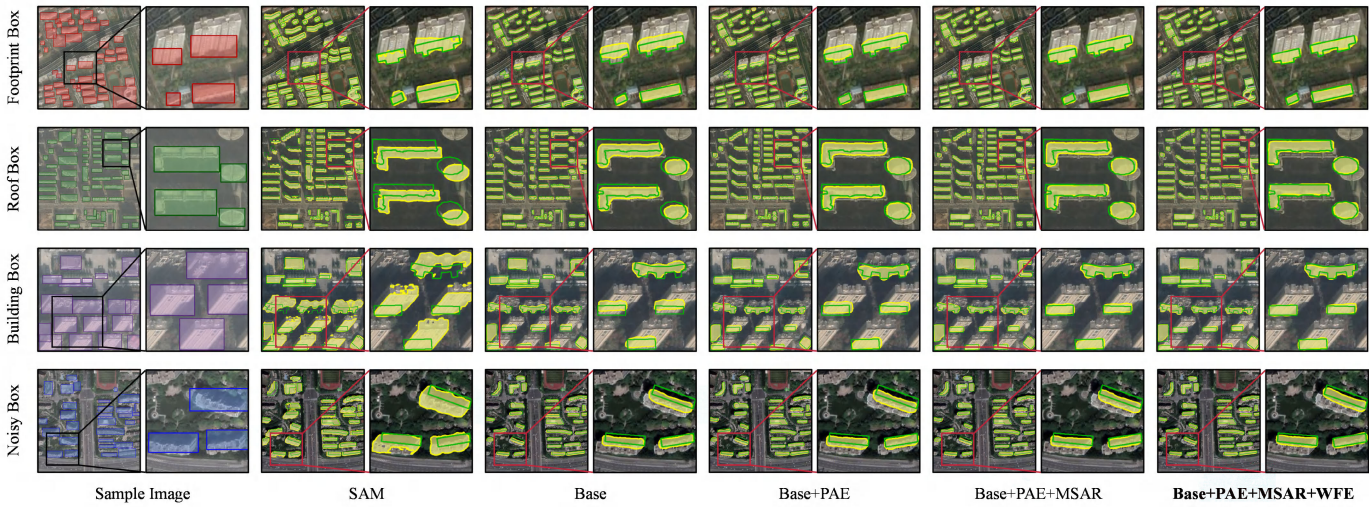


Fig. 8. Visualizations of the qualitative results from ablation experiments on the BONAI dataset, where the predicted results and the ground truth are highlighted in yellow and green, respectively.

TABLE VII

ABLATION STUDY OF THE PAE, MSAR, AND WFE MODULES FOR BUILDING FOOTPRINT EXTRACTION ON THE BONAI DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Footprint box			Roof box			Building box			Noisy box		
	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall
Baseline	80.82	82.81	79.09	78.29	80.32	76.49	80.44	80.25	80.82	77.16	77.13	77.39
+ PAE module	82.90	82.57	83.39	78.79	78.39	79.34	81.04	81.20	81.09	77.35	77.26	77.69
+ MSAR module	84.44	84.20	84.84	79.58	79.11	80.17	82.69	82.33	83.25	78.62	78.45	79.02
+ WFE module	**85.83**	**85.64**	**86.18**	**80.88**	**80.31**	**81.59**	**83.71**	**83.37**	**84.29**	**79.52**	**79.66**	**79.63**

TABLE VIII

ABLATION STUDY OF THE PAE, MSAR, AND WFE MODULES FOR OFFSET PREDICTION ON THE BONAI DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Footprint box			Roof box			Building box			Noisy box		
	aVE	aLE	aAE	aVE	aLE	aAE	aVE	aLE	aAE	aVE	aLE	aAE
Baseline	5.95	4.56	0.27	6.62	5.23	0.28	5.99	4.58	0.27	6.22	4.76	0.28
+ PAE module	5.92	4.55	0.27	6.59	5.15	0.29	5.99	4.61	0.28	6.24	4.85	0.28
+ MSAR module	5.42	4.24	0.25	6.05	4.79	0.26	5.47	4.18	0.26	5.83	4.53	0.27
+ WFE module	**5.09**	**4.05**	**0.23**	**5.58**	**4.48**	**0.24**	**5.05**	**3.96**	**0.23**	**5.37**	**4.25**	**0.24**

TABLE IX

ABLATION STUDY OF MSAR MODULE PARAMETERS ON THE BONAI DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Factor	Setting	Footprint box		Roof box		Building box		Noisy box	
		F1-score	aVE	F1-score	aVE	F1-score	aVE	F1-score	aVE
Dilation rates	[1, 1, 1, 1]	85.34	5.29	80.63	5.79	83.22	5.28	79.13	5.57
	[1, 2, 3, 4]	**85.83**	**5.09**	**80.88**	**5.58**	**83.71**	**5.05**	**79.52**	**5.37**
	[1, 3, 5, 7]	85.58	5.13	80.69	5.65	83.47	5.10	79.32	5.40
Offset scales	[1, 1, 1, 1]	85.36	5.17	80.72	5.67	83.33	5.14	79.27	5.43
	[0.5, 1, 2, 4]	**85.83**	**5.09**	**80.88**	**5.58**	**83.71**	**5.05**	**79.52**	**5.37**
	[1, 2, 4, 8]	85.79	5.11	80.84	5.60	83.60	5.08	79.42	5.38

TABLE X

ABLATION STUDY OF PRT STRATEGY SAMPLING RATIOS ON THE BONAI DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED. F, R, B, AND N DENOTE FOOTPRINT, ROOF, BUILDING, AND NOISY BOX PROMPTS, RESPECTIVELY

Sampling ratio (F : R : B : N)	Footprint box		Roof box		Building box		Noisy box	
	F1-score	aVE	F1-score	aVE	F1-score	aVE	F1-score	aVE
[2 : 1 : 1 : 1]	**85.90**	**4.99**	80.87	5.60	83.69	5.06	79.21	5.40
[1 : 2 : 1 : 1]	85.16	5.17	**81.42**	**5.39**	83.68	5.06	<u>79.51</u>	<u>5.32</u>
[1 : 1 : 2 : 1]	85.32	5.14	80.86	5.59	**84.00**	**4.95**	79.14	5.35
[1 : 1 : 1 : 2]	85.12	5.12	80.81	5.61	83.53	5.07	79.50	**5.31**
[1 : 1 : 1 : 1]	<u>85.83</u>	<u>5.09</u>	<u>80.88</u>	<u>5.58</u>	<u>83.71</u>	<u>5.05</u>	**79.52**	5.37

with error bars representing the standard deviation. The setting without the PRT strategy denotes that only ground-truth building boxes are used as prompts during training, without introducing additional prompt types. The results demonstrate that the PRT strategy substantially enhances footprint segmentation stability under diverse prompt conditions, yielding mean $F1$-score improvements of 8.63%, 14.96%, and 7.53% on the BONAI dataset with footprint box, roof box, and noisy box prompts, respectively. Consistent improvements

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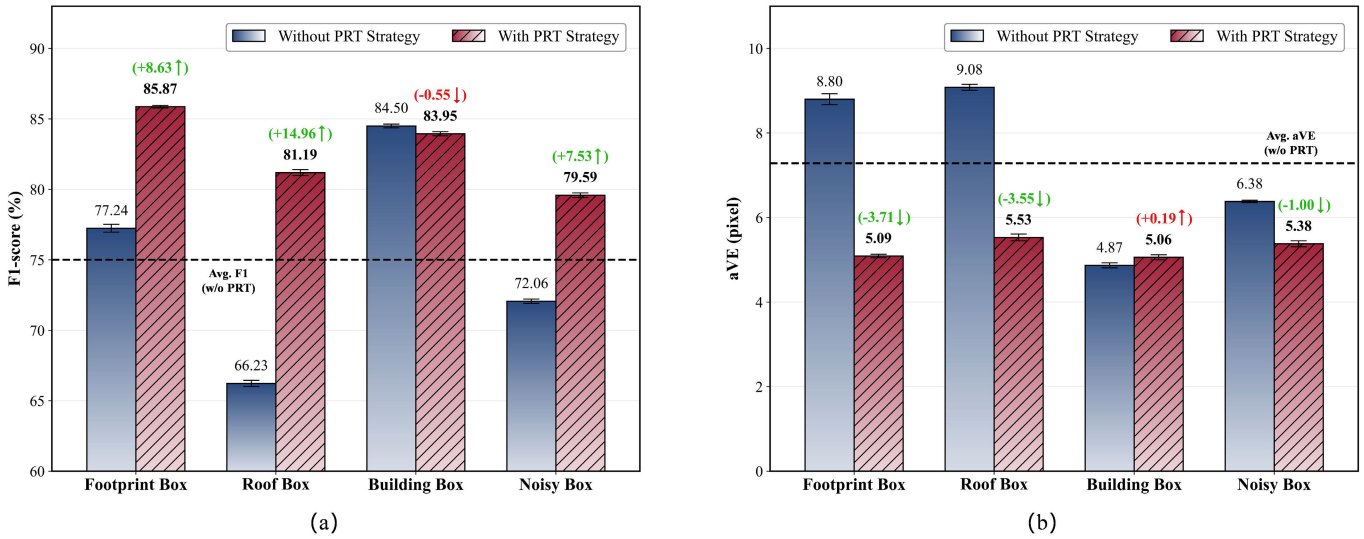


Fig. 9. Ablation study of the PRT strategy on the BONAI dataset for (a) footprint extraction and (b) offset prediction under diverse prompt conditions. Results are averaged over five random seeds, with error bars representing the standard deviation. The black dashed line indicates the average performance without PRT, while green and red values denote performance improvements and degradations, respectively.

in offset prediction accuracy are also shown in Fig. 9(b). It is worth noting that, compared to the baseline setting where training and inference prompts are identical (i.e., ground-truth building boxes), the PRT strategy incurs a slight performance drop of 0.55% in $F1$-score. Nevertheless, this tradeoff yields substantial gains in robustness across diverse prompt variations. Furthermore, we analyze the impact of different sampling ratios in the PRT strategy on model performance, as shown in Table X. Compared with equal-probability sampling [1:1:1:1], biasing training toward specific prompt types (e.g., [2:1:1:1], [1:2:1:1], and [1:1:2:1]) yields improvements for those prompt types but reduces robustness to others. In contrast, over-sampling noisy boxes [1:1:1:2] degrades accuracy across all prompt types. Therefore, we adopt the [1:1:1:1] setting, as it achieves the best tradeoff between overall robustness and accuracy.

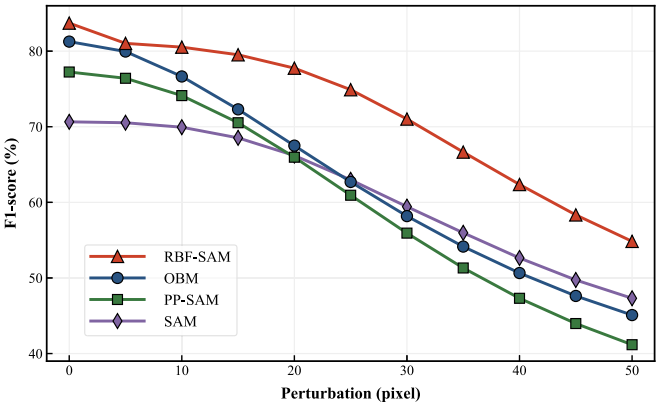


Fig. 10. Robustness boundary analysis on the BONAI dataset. $F1$-scores of different methods are reported under building box prompts with perturbation magnitudes ranging from 0 to 50 pixels.

D. Influence of Prompt Perturbations

To further investigate the robustness boundary of different models, we conduct experiments on the BONAI dataset by randomly perturbing the ground-truth building box prompts within a maximum range of 0–50 pixels. As shown in Fig. 10, RBF-SAM consistently outperforms SAM, PP-SAM, and OBM across all perturbation levels while exhibiting a slower degradation trend under increasing perturbations. Notably, the performance of existing methods becomes less satisfactory when the perturbation exceeds 15 pixels, with their $F1$-scores dropping below 70%. In contrast, RBF-SAM still maintains an $F1$-score of 71.02% under a 30-pixel prompt deviation, comparable to the performance of the prompt-robust method PP-SAM (70.53%) and the task-tailored method OBM (72.30%) under only a 15-pixel perturbation. These findings further confirm the superior prompt robustness of RBF-SAM.

E. Influence of Different Prompts on Model Predictions

We analyze the model's prediction behavior under different prompt conditions on the BONAI dataset. Specifically, under the joint influence of prompt spatial constraints and offset correction, we observe that the model's inference process can be broadly characterized by two main modes. When building box, roof box, or noisy box prompts are provided, the model tends to first localize the roof region and then relies on offsets to map it to the corresponding footprint, referred to as the roof-to-footprint mapping mode (Mode 1 in Fig. 11). As a result, the final prediction quality depends heavily on the accuracy of offset prediction. Since the building box prompt implicitly encodes more accurate offset information, it outperforms both roof box and noisy box prompts in offset prediction and footprint extraction, as reported in Tables IV and V.

By contrast, when footprint box prompts are provided, the inference behavior becomes more complex. While Mode 1 remains the primary mechanism, the strong spatial constraints imposed by the prompt frequently lead to a secondary behavior, known as the local refinement mode (Mode 2 in Fig. 11). In this mode, the spatial constraints reduce the model's reliance on long-range offset prediction. Consequently, the

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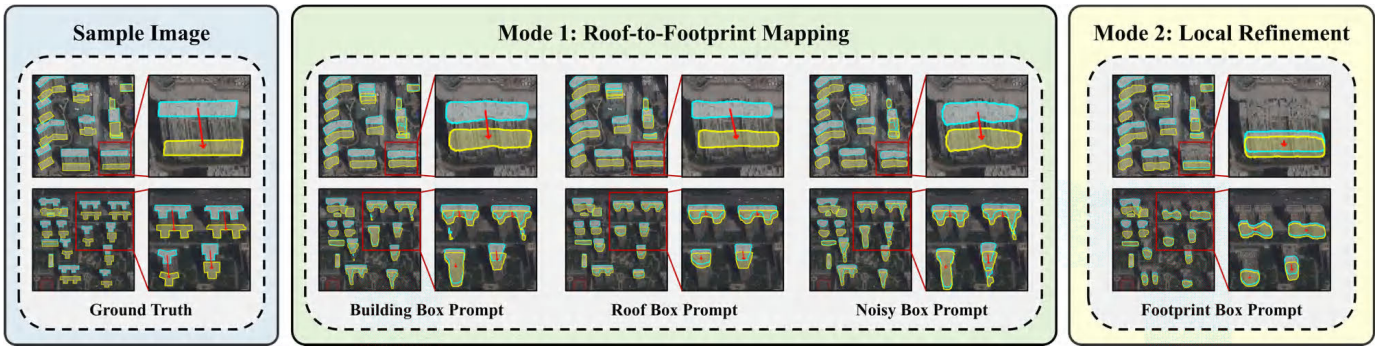


Fig. 11. Analysis of the model's prediction behavior under different prompt conditions. Mode 1: Roof-to-footprint mapping, relying on large offsets for spatial mapping and serving as the primary mechanism across prompt types. Mode 2: Local refinement, a secondary behavior under footprint box prompts, characterized by strong spatial constraints and small-magnitude offsets. Cyan and yellow contours denote the roof and footprint, respectively, with offsets indicated by red arrows.

model tends to predict internal structures closer to the footprint and applies only small-magnitude offsets for local refinement. Although the limited offset magnitude leads to lower offset prediction accuracy, localizing the target closer to the footprint ultimately enhances the overall extraction performance, as reported in Table IV.

These observations suggest that the model adjusts its inference strategy based on the spatial priors provided by the prompts. In summary, this analysis reveals the inference behaviors of the prompt–offset framework under varying prompt conditions, which may offer useful insights for the design of future SAM-based off-nadir building extraction networks.

F. Current Limitations and Future Directions

While RBF-SAM demonstrates strong robustness across diverse prompt conditions, several challenges remain. First, the framework relies on interactive prompts (particularly dense box prompts) to obtain reliable extraction results, which limits its scalability in large-scale remote sensing applications. Second, although the proposed method achieves promising performance, the introduced MSAR and WFE modules incur additional computational overhead. Finally, in extremely complex scenarios, such as dense building areas, the accuracy of the predicted footprints may still be compromised.

In future work, we will explore sparse point-based prompting [73], [74] or automatic prompt generation methods [24], [25] to reduce manual intervention costs, and investigate lightweight strategies [36], [75] to improve computational efficiency. Moreover, integrating advanced feature representations [48] and enriched multimodal information [60] will be considered to enhance the model's performance, alongside further investigation under varying off-nadir imaging conditions.

VI. Conclusion

In this article, we present the RBF-SAM, a framework tailored for off-nadir remote sensing imagery. To the best of our knowledge, RBF-SAM is the first prompt-driven building footprint extraction approach that systematically addresses robustness across varying prompt qualities and types. Built upon SAM, RBF-SAM integrates the PAE module and the MSAR module to mitigate sensitivity to inaccurate prompts. Moreover, we introduce the WFE module to enrich feature representations in the frequency domain, thereby boosting the accuracy of both boundary delineation and offset prediction.

Comprehensive experiments on the BONAI and OmniCity-view3 datasets demonstrate that RBF-SAM consistently outperforms existing methods under a wide range of prompt conditions. Furthermore, evaluations on the Huizhou test set validate the strong cross-region generalization capability of the proposed approach. Future work will explore more automatic and lightweight frameworks with multimodal inputs for robust building extraction in complex off-nadir scenarios.

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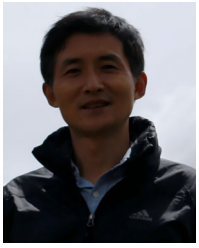
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